❶ Read: KBo 44.52, an orphan (CTH 470)

1′ x-a-ḫi-
2′ NINDA.SIG x x
3′ EN SÍSKUR-ma AN
4′ LÚSIMUG-ia ku-e x
5′ LÚḪAL ša-ra-a da-a-i
6′ xḪLA-ma-aš-ša-an ku-i-e-eš A-NA GIŠBANŠUR
7′ x da-aš-ke-ez-zi na-at-kán ḫa-aš-ši-i zi-Ø-Ø-Ø-Ø
8′ -in-ni zi-ik-ke-ez-zi EN SÍSKUR-ma
9′ na-an-kán EN SÍSKUR A-NA DIŠTAR ši-pa-Ø-Ø
10′ šu-up-pa ḫu-u-i-šu da-an-zi UZUGABA
11′ ØBANŠUR AD.KID da-ga-a-an da-a-i še-ra-Ø-Ø-Ø
12′ Ø-NA GIŠBANŠUR še-er da-a-i še-er ar-Ø
13′ -zi na-at-kán an-da ú-
14′ nu-uš-ši-kán LÚAZU x
15′ ØBANŠUR da-a-i na-an-ša-Ø
16′ -ma-aš-ša-Ø

function verb realia agent, deity place animal, body, nature abstract adjective foreign unclear

❷ Count

Each word becomes several features, its form and its sign sequences:

da-a-i

→ word **tai**	0.09
→ signs *da-a-i*	0.09
→ signs *da-a*	0.08
→ signs *a-i*	0.08

A feature weighs more if it is frequent here and rare in the corpus:

$$x_f \propto (1 + \log tf_f) \cdot idf_f$$

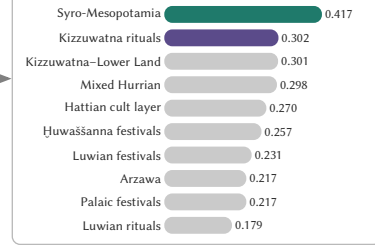
idf of **tai** 1.9; of the sign *an* 1.1

The fragment is a vector $\mathbf{x}$ with 160 of 20,166 features.

❸ Compare

Each tradition is the average vector $\mathbf{c}_t$ of its manuscripts. Similarity is the cosine:

$$s_t = \cos(\mathbf{x}, \mathbf{c}_t) = \sum_f x_f c_{t,f}$$



❹ Decide

The lead of the nearest tradition over the runner-up:

$$m = s_1 - s_2 = 0.42 - 0.30 = 0.12$$

Placed only if $m \geq \tau = 0.028$, the lead at which 80% of test fragments are placed correctly; otherwise left unassigned.

⇒ **Syro-Mesopotamia**

❺ Explain

The lead splits exactly over the features:

$$m = \sum_f x_f (c_{1,f} - c_{2,f})$$

feature	x_f	$c_{1,f}$	$c_{2,f}$	share ×10³
AZU	0.07	0.13	0.02	+7.8
AZU	0.07	0.12	0.02	+6.8
tai	0.09	0.13	0.05	+6.8
SÍSKUR	0.13	0.08	0.03	+6.6
LÚ-AZU	0.07	0.11	0.01	+6.5
da-a-i	0.09	0.10	0.04	+5.1
ga	0.05	0.02	0.06	−1.6
ke	0.08	0.03	0.05	−1.6

c_1 Syro-Mesopotamia, c_2 Kizzuwatna rituals; bold: word form, italic: signs.

❻ Check by hand

1′ x-a-ḫi-
2′ NINDA.SIG x x
3′ EN SÍSKUR-ma AN
4′ LÚSIMUG-ia ku-e x
5′ LÚḪAL ša-ra-a da-a-i
6′ xḪLA-ma-aš-ša-an ku-i-e-eš A-NA GIŠBANŠUR
7′ x da-aš-ke-ez-zi na-at-kán ḫa-aš-ši-i zi-Ø-Ø-Ø-Ø
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15′ ØBANŠUR da-a-i na-an-ša-Ø
16′ -ma-aš-ša-Ø

Darker: larger share of the lead.

AZU, **tai**, **BANŠUR** decide most. Of the five nearest single manuscripts, 3 are CTH 701, starting with KBo 59.165: leads for the philologist.